frontiers

Supplementary Material

1 SUPPLEMENTARY METHODS AND MATERIALS

2 SUPPLEMENTARY TABLES AND FIGURES

2.1 Tables

Table S1. Comparison of pre- and post-optimization torque prediction fits for BPAs in flexor and extensor configurations on the pinned and biomimetic knees. GoF metrics are RMSE ($N \cdot m$), FVU, and maximum residual ($N \cdot m$).

Configuration		ϕ	$\mathbf{l}_{rest}, cm$	**Pre-optimization**			**Optimized**		
				RMSE	**FVU**	**Max. Residual**	**RMSE**	**FVU**	**Max. Residual**
Flexor	**Pinned**	10	48.5	4.871	0.723	9.865	1.608	0.095	3.242
		10	45.7	3.024	0.163	6.753	1.915	0.072	4.695
	Biomimetic	10	41.5	7.448	5.557	13.308	0.680	0.052	1.055
		20	41.5	14.803	3.366	23.715	2.868	0.126	3.550
Extensor	**Pinned**	10	41.5	4.520	3.972	11.741	2.042	0.844	3.588
		10	45.7	1.999	0.892	4.609	0.896	0.179	1.846
		10	48	0.845	0.232	1.951	0.682	0.151	2.068
	Biomimetic	10	51.8	3.032	4.486	5.587	1.461	1.041	2.755

Table S2. Muscle origin and insertion locations for different models. Origin is in femur reference frame, insertion is in the tibia reference frame.

Muscle	Origin/Insertion	Model	X (m)	Y (m)	Z (m)
Flexor	origin	Pinned knee	-0.075	0.100	0.0328
Flexor	insertion	Pinned knee	-0.05011	-0.045	0.0326
Extensor	origin	Pinned knee	0.030	-0.050	0
Extensor	insertion	Pinned knee	0.0425	-0.076	0
Bifemsh	origin	Gait2392	0.005	-0.211	0.023
Bifemsh	insertion	Gait2392	-0.03	-0.036	0.029
Vastus int.	origin	Gait2392	0.029	-0.192	0.031
Vastus int.	insertion	Gait2392	moving	moving	0.0018
Flexor, $10\,\mathrm{mm}$	origin	Biomimetic knee	-0.050	-0.045	0.0328
Flexor, $10\,\mathrm{mm}$	insertion	Biomimetic knee	-0.0279	-0.046	0.0328
Extensor, $10\,\mathrm{mm}$	origin	Biomimetic knee	0.040	0.035	0
Extensor, $10\,\mathrm{mm}$	insertion	Biomimetic knee	0.021	-0.072	0
Flexor, $20\,\mathrm{mm}$	origin	Biomimetic knee	-0.050	-0.045	0.0328
Flexor, $20\,\mathrm{mm}$	insertion	Biomimetic knee	-0.0279	-0.046	0.0328

2.2 Figures

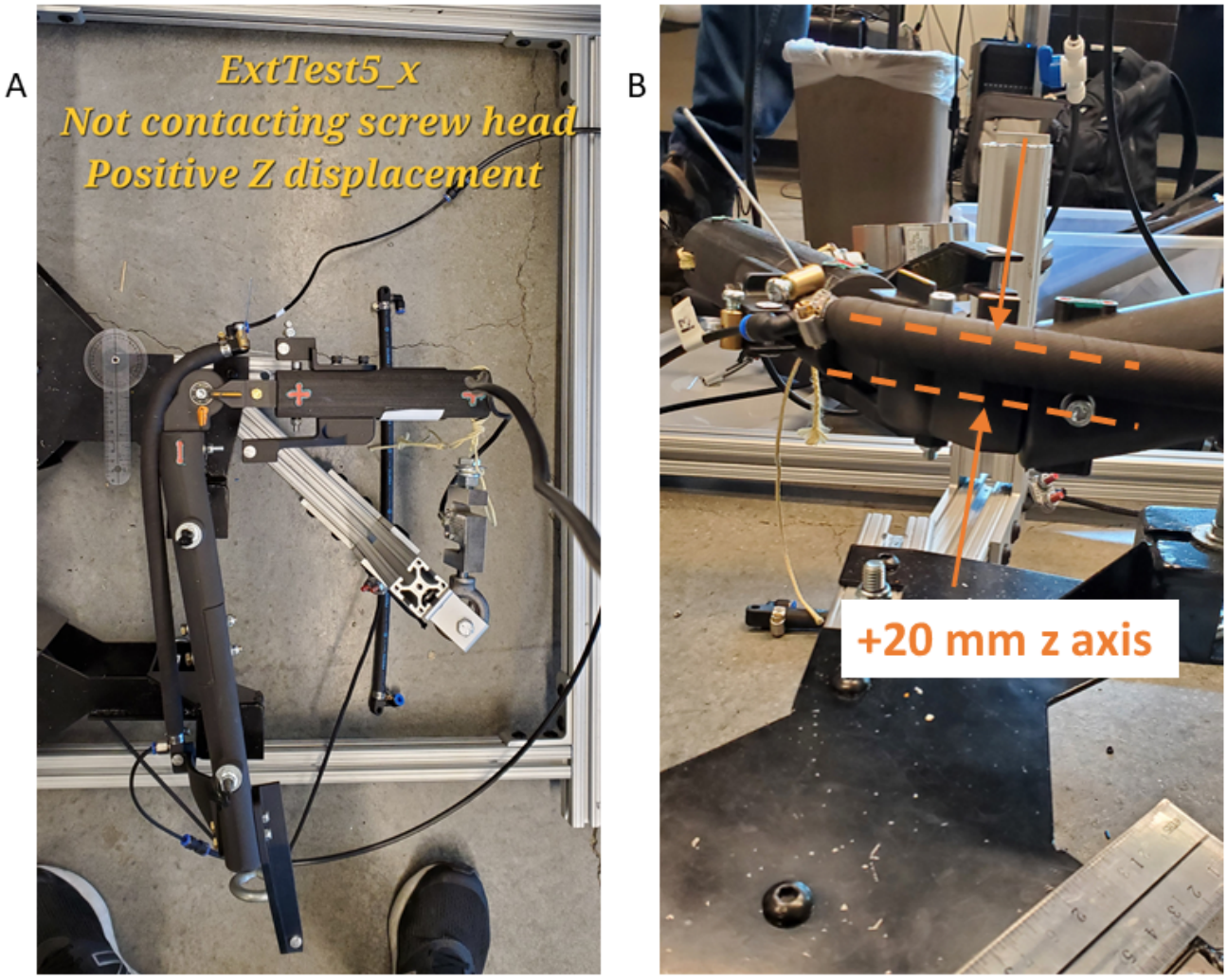


Figure S1: During extensor tests with the pinned knee using a tendon it was noticed that the actuator pathway was modeled to be anterior of the socket head cap screw that joins the femur body to the knee assembly. This slippage affects the muscle path and the moment arm. It was determined this caused a $\pm 20\,\mathrm{mm}\ \hat{z}$ axis displacement. Test assembly is shown with **(A)** top down view and **(B)** side view.

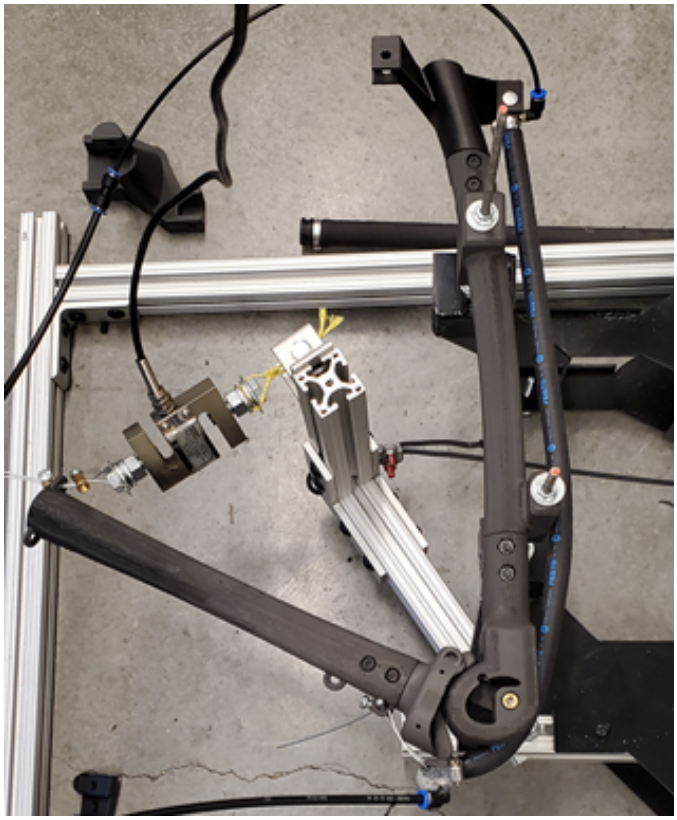


Figure S2: Biomimetic 2-DoF robot leg installed in the extensor test configuration. Note how the BPA passes through the femoral condyle.